

## Heat Transfer

Final Lab Report  
Unit Operations Lab 2  
February 19, 2020

Group 2  
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## 1. Executive Summary

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The purpose of this experiment was to calculate the heat transfer coefficient between a fluid (water) and the inside of an agitated tank. Another objective was to examine the dependence of the heat transfer coefficient on impeller speed, fluid properties, and the presence of baffles. In Experiment A, a total of five runs were completed using three different impeller speeds as well as two recycle flow rates. In Experiment C, a total of four runs were completed by varying two impeller speeds and varying two initial steam conditions. Finally, data was collected and used to calculate heat transfer coefficients using equations from theory.

In Experiment A “Wall HTC for the Unbaffled Agitated Tank,” three different impeller speeds were varied in order to examine effects on heat transfer coefficient. Impeller speeds of 610, 700, and 785 rpm were used in three trials. Volumetric flow rate of cooling water in the cooling coil, cooling water from the heat exchanger, and recirculation water flow rate were recorded from the rotameter at 0.3 gpm, 0.775 gpm, and 0.143 gpm, respectively. Next, thermocouple temperatures were recorded in each trial for the water in the vessel, cooling water inlet and outlet from the cooling coil, process water inlet and outlet from the heat exchanger, and cooling water inlet and outlet from the steam trap. Lastly, steam condensate was timed and collected with a graduated cylinder to give the following volumetric flow rates with corresponding impeller speeds: 4.2373 mL/s (610 rpm); 5.6878 mL/s (700 rpm); 6.2562 mL/s (785 rpm).

In the last two runs of Experiment A, water recirculation flow rate was varied from 0.292 gpm to 0.649 gpm at a constant impeller speed of 710 rpm. Once again, thermocouple temperatures and steam condensate were collected in the same fashion as the three impeller speed trials. Steam condensate was calculated to be 3.040 mL/s at a recycle flow rate of 0.292 gpm and 5.2947 mL/s at a recycle flow rate of 0.649 gpm. Cooling water in the cooling coil and the heat exchanger were kept constant at 0.3 gpm and 0.776 gpm, respectively.

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In Experiment C “Unsteady State Wall HTC for the Unbaffled Agitated Tank,” four different runs were completed using two different initial steam conditions (shut-off, turned on) and two impeller speeds of 730 and 900 rpm. The recirculation pump and cooling coil were not used in this experiment. From here, the cooling water inlet and outlet temperature was recorded. Volume flow rate was calculated by timing the collection of condensate in a graduated cylinder and recording the data before the system achieved steady-state, contrary to Experiment A. A maximum volumetric flow rate of 29.07 mL/s was obtained at an impeller speed of 900 rpm and vessel water temperature of 37.79°C. A minimum volumetric flow rate of 3.22 mL/s was obtained at an impeller speed of 730 rpm and vessel water temperature of 82.22°C.

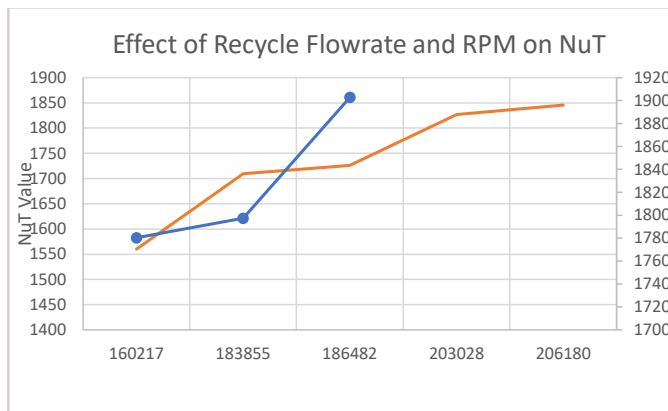


Figure 1: Summary of NuT increase trends with Flowrate and RPM

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## 2. Introduction

For the heat transfer lab, the main objective is to measure the heat transfer coefficient between the water and the vessel wall. It will also be important to study the dependence of the heat transfer coefficient on the impeller speed, fluid properties, and use of mechanical baffles. All of these are very dependent on temperature. The experimentally measured heat transfer coefficient will also be compared to values of other investigators.

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The apparatus setup for this experiment included a steam jacketed vessel, pump for circulating the cooling water, a heat exchanger, and many other components. The vessel has an internal cooling coil and dial thermometers are located at the inlet and outlet of the coil. The water from the tank is continuously withdrawn from the bottom and circulated through the heat exchanger. The temperature of the recirculated tank water is measured at the inlet and outlet of the heat exchanger. There are three different experiments that can be performed for this lab: Wall HTC for the unbaffled agitated tank, Wall HTC for the baffled agitated tank, and Unsteady state wall HTC for the unbaffled agitated tank. Experiment A and C were performed for this lab.

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Agitated tanks are very commonly used in industrial applications. They are used in industries that process products in the chemical, food, pharmaceutical, and cosmetic industries [1]. They are used for the mixing of liquids, promote reactions of chemical substances, and to increase heat transfer. These tanks are used for such a variety of processes and that is why it is such an important piece of equipment.

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## 3. Theory

In chemical engineering heat transport is an extremely important concept. The understanding of this concept is key to designing proper equipment and processes. When analyzing a system involving a vessel that is well stirred by an impeller, there is heat transfer between the vessel wall, which is heated by a steam jacket, and the fluid inside. As with any transport problem, it begins with the basic governing equation shown below (Equation 1).

$$IN - OUT + GENERATION = ACCUMULATION$$

Equation 1

For thermal conductivity, a simple equation to use is Fourier's Law (Equation 2).

$$\frac{dq}{dA} = -k \frac{\partial T}{\partial n}$$

Equation 2

Where  $A$  is the area of the isothermal surface,  $n$  is the normal distance from the surface,  $q$  is the heat flow across a surface in the normal direction,  $T$  is temperature, and  $k$  is the proportionality constant [2] [3].

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One important concept to develop is the overall heat balance on the system. In general, the heat balance can be written from Equation 1 as shown below.

$$Heat\ IN - Heat\ OUT = ACCUMULATION\ of\ Heat\ in\ Fluid$$

This can be written from heat balances shown below.

$$Heat\ IN = q_{in} = \dot{m}(H_b - H_a) = \dot{m}C_p\Delta T$$

Equation 3

Where  $q_{in}$  is the heat transfer into the system,  $\dot{m}$  is the mass flow rate,  $H_a$  and  $H_b$  are the specific enthalpies per unit mass in the entrance and exit respectively,  $C_p$  is the heat capacity, and  $\Delta T$  is the change in temperature between the entrance and exit [4].

When considering heat transfer across a wall such as the vessel used in this experiment it is more appropriate to write Equation 3 in terms of the overall heat transfer coefficient as shown in Equation 4.

$$Q = U_i A_i \Delta T \quad \text{Equation 4}$$

Where  $Q$  is the heat transfer into the system,  $U_i$  is the overall heat transfer coefficient of the interface, and  $A_i$  is the surface area of the interface [5]. Note that if it assumed that the wall thickness is small compared to the vessel diameter (which was the case in this experiment)  $U_i$  can be estimated using the following formula.

$$U_i = \frac{1}{\left(\frac{1}{h_i}\right) + \left(\frac{1}{h_o}\right)} \quad \text{Equation 5}$$

Where  $h_i$  and  $h_o$  are the heat transfer coefficients of the inside and outside of the vessel wall respectively [5].

### Experiment A

For Experiment A, the heat balance is given below.

$$Q_{in} = Q_{hx} + Q_c + Q_{surr} \quad \text{Equation 6}$$

Where  $Q_{in}$  is the heat transfer into the system,  $Q_{hx}$  is the heat removed by the heat exchanger,  $Q_c$  is the heat removed by the cooling coil,  $Q_{surr}$  is the heat lost to the surroundings. Since this experiment is assumed to be at steady state, the accumulation term in Equation 1 is zero.

A few important terms used in calculations are defined below. These are dimensionless groups often used in many calculations involved transport phenomena.

$$\text{Tank Nusselt Number} = Nu_T = \frac{h_i D_T}{k} \quad \text{Equation 7}$$

$$\text{Impeller Reynolds Number} = Re_{imp} = \frac{D_I^2 n \rho}{\mu} \quad \text{Equation 8}$$

$$\text{Fluid Prandtl Number} = Pr = \frac{C_p \mu}{k} \quad \text{Equation 9}$$

Where

- $D_T$  is the vessel diameter
- $k$  is the thermal conductivity
- $\mu$  is the viscosity of the fluid
- $\rho$  is the density of the fluid
- $D_I$  is the impeller diameter
- $n$  is the revolutions per unit time of the impeller
- All other terms are defined above

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The terms are related through the following formula [6].

$$Nu_T = 0.36(Re_{imp})^{\frac{2}{3}}(Pr)^{\frac{1}{3}} \quad \text{Equation 1}$$

The heat transfer coefficient on the outside of the wall (the steam side) can be estimated from the correlation below.

$$\frac{h_o L_T}{k} = 0.925 \left( \frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{\frac{1}{3}} \quad \text{Equation 11}$$

Where

- $\Gamma$  is the mass rate of steam condensation per wetted perimeter given by  $\frac{\dot{m}_{st}}{\pi D_{T,o}}$
- $\dot{m}_{st}$  is the mass flow rate of steam condensate
- $D_{T,o}$  is the vessel tank outer diameter
- $L_T$  is the vessel length
- $g$  is gravitational acceleration
- All other terms are defined above

Note this equation (Equation 11) can only be used if  $Re_{film} = \frac{4\Gamma}{\mu}$  is less than 2100 [5].

#### Experiment C

In this experiment, the vessel temperature is a function of time and therefore, not at steady state. The cooling water is shut off to the cooling coil and the process fluid is not flowing through the heat exchanger. This means the energy out of the system is essentially zero, which gives the following energy balance.

$$UA\Delta T = m_v C_p \frac{dT}{d\theta} \quad \text{Equation 12}$$

Where  $m_v$  is the mass of the fluid in the vessel,  $\theta$  is the time, and all other terms are defined above. The overall heat transfer coefficient can be estimated Equation 12 by assuming the derivative term can be estimated using small changes (ie.  $\frac{\Delta T}{\Delta \theta}$ ).

#### 4. Results

This lab sought to measure the heat transfer between a continuous stirred tank. A comparison of the unsteady state heat transfer coefficient and the steady state heat transfer coefficient was chosen. Various temperatures were measured along the CST as well as flow rates for the recycled process water, and cooling water through the coils and the exterior heat exchanger.

Table #1: Conclusive Results from Steady State heat transfer

| Case | RPM | Recycle FlowRate<br>(gpm) | Q13<br>(kJ) | U<br>(W/m <sup>2</sup> K) | NuT_corr | NuT_calc |
|------|-----|---------------------------|-------------|---------------------------|----------|----------|
| 1    | 610 | 0.143                     | 6.10        | 0.157                     | 1624     | 1560     |
| 2    | 700 | 0.121                     | 5.37        | 0.136                     | 1780     | 1710     |
| 3    | 773 | 0.143                     | 5.70        | 0.129                     | 1903     | 1827     |
| 4    | 785 | 0.242                     | 4.51        | 0.109                     | 1922     | 1846     |
| 5    | 710 | 0.2915                    | 10.39       | 0.250                     | 1797     | 1726     |
| 6    | 710 | 0.649                     | 3.20        | 0.075                     | 1797     | 1726     |

**Commented [DL14]:** What's the assumption of this empirical equation?  
(Inner wall, unbaffled, turbulent flow and Newtonian fluid.)

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The purpose of this experiment was to calculate the heat transfer coefficient between a fluid (water) and the inside of an agitated tank.  
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By looking at table #1. The steady state heat transfer values the heat duty decreased with increasing RPM and that the heat transfer coefficient decreased as well. The heat duty however increased with increasing recycled water flowrate and the heat transfer coefficient decreased. The correlated equation for the heat transfer coefficient in the form of the Nusselt number provides a relatively good estimate for the true calculated Nusselt number.

Using equations 7, 8, and 9, the Nusselt Number, Reynolds number, and Prandtl number were calculated for 3 different impeller speeds while keeping the recycle flowrate at a constant flowrate. By observing a change in the impeller speed, a trend in the heat transfer coefficient can be observed by a better or worse mixing of the fluid.

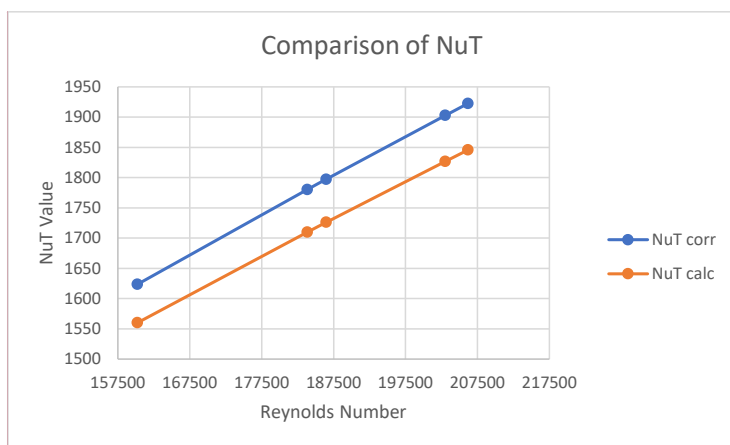


Figure 2: Steady State NuT comparisons as function of Reynolds number

Trends for the steady state heat transfer value as a function of Reynolds number can be found in figure 2. A correlation equation (10) was used to compare the calculated heat transfer value compared to the expected heat transfer value. The recycle water flowrate was increased for 3 different volumetric flows while keeping the RPM of the motor constant to observe the heat transfer values. Data for increasing Reynolds number because of the increase in RPM can be found in tables 2-8 in Appendix I.

As shown in table 1, the heat duty of the continuously stirred tank increased with increasing Reynolds number. The overall heat transfer coefficient increased as well. When the recycle flowrate increased, the duty and the overall heat transfer coefficient were similarly increased.

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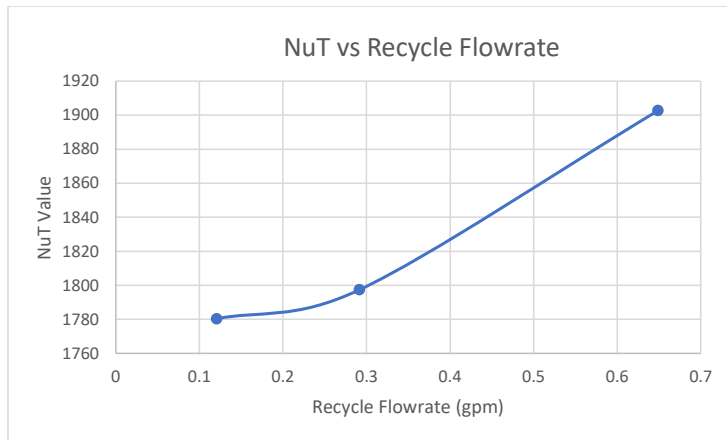


Figure 3: Steady State NuT as function of recycle flowrate

Trends for the steady state heat transfer value as a function of recycle process water flowrate can be also graphed and can be found in figure 3. A correlation equation (10) was used to compare the calculated heat transfer value compared to the expected heat transfer value. There is a difference in the expected versus the measured value because of the correlations assumptions when modeling it. Data for increasing recycle process water can be found in tables 9-14 in appendix.

Unsteady state calculation was performed with the same equations as with the steady state. The measurements however were taken in temperature intervals of 5 Fahrenheit. This way, a trend curve would show the rise or decline of heat transfer as time progressed. Since different time intervals have individual Reynolds and Prandtl numbers, figures 4 and 5 can be expressed as functions of Reynolds number and still be function of time since Reynolds number is dependent on time itself.

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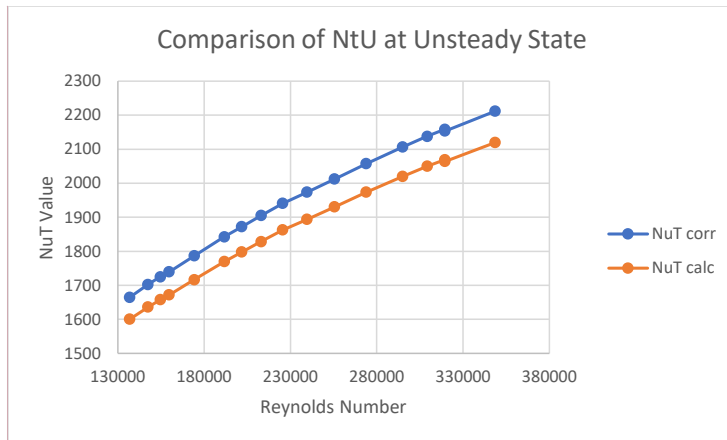


Figure 4: Unsteady state NuT comparisons at low RPM

Figure 4 show the trend in heat transfer as a function of Reynolds number at unsteady state conditions. For comparison of the unsteady state trend, two different RPM of the motor were chosen. Once more, the same correlation as used in steady state was used again to compare the measured NuT value with the expected NuT value. Figure 4 demonstrates unsteady state trends at a lower RPM value of the motor than figure 5.

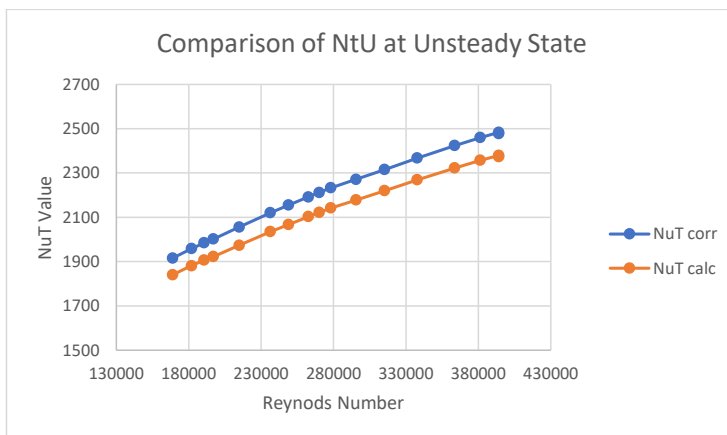


Figure 5: Unsteady State NuT comparison for high RPM

In both figure 4 and 5, two sets of data are available per experimental run. The final graphed data was the average value of the two measured and calculated values.

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## 5. Discussion

From figures 2, 3, and 4, we can see that the Nusselt number correlation given by equation 10 appears to fit our experimental results well because the calculated and correlated Nu values appear to be correlated. While there is a discrepancy between the actual data and the calculated data, this deviation was expected since the correlation equations takes certain assumptions that aren't true for our case.

Looking at figure 2, the heat transfer value increased with increasing Reynolds number. This means that with an increase in RPM, the quality of mixing did contribute to the overall heat exchange between the cooling and the process fluid itself. Ideally, better mixing would result in an even great increase in heat transfer.

Looking at figure 3, the heat transfer value increased once again when the RPM of the motor remained fixed and the recycle flowrate of the process water increased. This means that the increase in the recycle water provided a better opportunity for heat transfer to take place. More water was able to exchange heat faster.

Looking at figure 4 and 5, the unsteady state heat transfer trend can be observed. Since measurements were taken in time (and consequentially temperature) intervals, we can see that the heat transfer value increased gradually as time went on. Since the Reynolds and Prandtl number are themselves function of time because each data point collected would have a different value, the plot against Reynolds number is akin to plotting against time.

Two different RPM were used for comparison of unsteady state conditions. It can be seen that the unsteady state at higher RPM values of the motor achieved higher heat transfer values than the lower RPM. This aligns with the conclusion taken from the steady state data measured and calculated. Higher Reynolds number flow result in better mixing allowing for more heat transfer. While the RPM of the motor increased, the trend in increasing heat transfer followed the same fashion. The heat transfer was higher, but it didn't mean the curve would become steeper.

From tables 17 and 18, we can also see that the Nusselt number rises with rising temperatures in non-steady state operation. The fact that the Nusselt number also rose with increasing recycle flow means that the recycle stream caused an increase in relative conductive heat transfer. This may be due to an increased mixing that was caused by the flow of the stream or it may have been caused by a decrease in the conductive heat transfer caused by a lower temperature in the tank that decreased the driving force for conductive transfer.

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**6. References**

- [1] "Industrial Agitator." *Wikipedia*, Wikimedia Foundation, 13 Nov. 2019, en.wikipedia.org/wiki/Industrial\_agitator.
- [2] R.B. Bird, W.E. Stewart, and E.N. Lightfoot, "Chapter 10: Shell Energy Balance and Temperature Distributions in Solids and Laminar Flow." *Transport Phenomena*, 2<sup>nd</sup> ed. John Wiley and Sons, Inc., 2002, pp. 290-320.
- [3] McCabe, Warren L., and Julian C. Smith. "Chapter 10: Heat Transfer by Conduction." *Unit Operations of Chemical Engineering*, 5th ed., McGraw Hill, 1956, pp. 289-306.
- [4] McCabe, Warren L., and Julian C. Smith. "Chapter 11: Principles of Heat Flow in Fluids." *Unit Operations of Chemical Engineering*, 5th ed., McGraw Hill, 1956, pp. 309-328.
- [5] "Heat Transfer to a Fluid in a Stirred Tank." ChE 382: Unit Operations Laboratory.
- [6] R. H. Perry and D. W. Green, Eds., *Perry's Chemical Engineer's Handbook*, 7<sup>th</sup> ed. McGraw-Hill, 1997.

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## 7. Appendix I: In-Lab Measurements

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Table 2: Steady State Results Summary

| Case | RPM | Recycle FlowRate (gpm) | Q13 (kJ) | U (W/m <sup>2</sup> K) | NuT_corr | NuT_calc |
|------|-----|------------------------|----------|------------------------|----------|----------|
| 1    | 610 | 0.143                  | 6.10     | 0.157                  | 1624     | 1560     |
| 2    | 700 | 0.121                  | 5.37     | 0.136                  | 1780     | 1710     |
| 3    | 773 | 0.143                  | 5.70     | 0.129                  | 1903     | 1827     |
| 4    | 785 | 0.242                  | 4.51     | 0.109                  | 1922     | 1846     |
| 5    | 710 | 0.2915                 | 10.39    | 0.250                  | 1797     | 1726     |
| 6    | 710 | 0.649                  | 3.20     | 0.075                  | 1797     | 1726     |

Table3: Lowest Impeller Speed inlab Measurements

|                        | Location      | Value  | Unit |      |                  |             |              |
|------------------------|---------------|--------|------|------|------------------|-------------|--------------|
| RPM of Impeller (np)   | Top           | 610    | rpm  |      |                  |             |              |
| FlowRates              |               |        |      |      |                  |             |              |
| CW in cooling coil     | Left          | 0.3    | gpm  |      | Steam Temp       | 100         | C            |
| Recirculation Water    | Right         | 0.143  | gpm  |      | Steam Condensate |             |              |
| CW to HEX              | Back          | 0.7755 | gpm  |      | Time (s)         | Volume (mL) | Vflow (mL/s) |
| Temperatures           |               |        |      |      | 20.06            | 85          | 4.2373       |
| WATER                  | On Vessel     | 154    | F C  | 67.8 |                  |             |              |
| CW IN                  | Back left     | 38     | F C  | 3.3  | Viscosity        | 0.0005      | kg/m*s       |
| CW OUT                 | Back Right    | 84     | F C  | 28.9 | Cp               | 4184        | J/smK        |
| PW before HEX          | Right bott    | 157.1  | F C  | 69.5 | Thermal K        | 0.66        | J/smK        |
| PW after HEX           | Right top     | 121.1  | F C  | 49.5 | (pg62)           |             |              |
| CW Temp IN steam trap  | Bottom        | 40     | F C  | 4.4  |                  |             |              |
| CW Temp OUT steam trap | Bellow Vessel | 48     | F C  | 8.9  |                  |             |              |

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Table4: Calculations for Lowest Impeller Speed

|                                 |             |
|---------------------------------|-------------|
| Nusselt Number (NuT)            | 1560.0      |
| Reynold Number (Reimp)          | 160217      |
| Prandtl Number (Pr)             | 3.17        |
| m_st (kg/s)                     | 2.54E-06    |
| Mass Rate per wet perim (Gamma) | 1.07E-06    |
| h0                              | 1.98E+05    |
| Qhx (J)                         | 4.085E+03   |
| Qcoils (J)                      | 2.02E+03    |
| Qsurr (J)                       | 0.00        |
| Q13 (kJ)                        | 6.10        |
| Overall Heat Trans (U)          | 0.157080259 |

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|                 |          |
|-----------------|----------|
| NuT correlation | 1.62E+03 |
| h0_wall         | 1.61E+05 |

Table5: Medium Impeller Speed InLab Measurements

|                        | Location      | Value | Unit |      |                  |             |              |
|------------------------|---------------|-------|------|------|------------------|-------------|--------------|
| RPM of Impeller (np)   | Top           | 785   | rpm  |      |                  |             |              |
| FlowRates              |               |       |      |      |                  |             |              |
| CW in cooling coil     | Left          | 0.3   | gpm  |      | Steam Temp       | 100         | C            |
| Recirculation Water    | Right         | 0.242 | gpm  |      | Steam Condensate |             |              |
| CW to HEX              | Back          | 0.775 | gpm  |      | Time (s)         | Volume (mL) | Vflow (mL/s) |
|                        |               | 5     |      |      | 10.15            | 63.5        | 6.2562       |
| Temperatures           |               |       |      |      |                  |             |              |
| WATER                  | On Vessel     | 150   | F C  | 65.6 |                  |             |              |
| CW IN                  | Back left     | 38    | F C  | 3.3  | Viscosity        | 0.0005      | kg/m*s       |
| CW OUT                 | Back Right    | 85    | F C  | 29.4 | Cp               | 4184        | J/smK        |
| PW before HEX          | Right bott    | 152.6 | F C  | 67.0 | Thermal K        | 0.66        | J/smK        |
| PW after HEX           | Right top     | 131   | F C  | 55.0 |                  | (pg62)      |              |
| CW Temp IN steam trap  | Bottom        | 41    | F C  | 5.0  |                  |             |              |
| CW Temp OUT steam trap | Bellow Vessel | 48    | F C  | 8.9  |                  |             |              |

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Table6: Calculations for Medium Impeller Speed

|                                 |           |
|---------------------------------|-----------|
| Nusselt Number (NuT)            | 1845.6    |
| Reynold Number (Reimp)          | 206180    |
| Prandtl Number (Pr)             | 3.17      |
|                                 |           |
| m_st (kg/s)                     | 3.75E-06  |
| Mass Rate per wet perim (Gamma) | 1.58E-06  |
| h0                              | 1.73E+05  |
| Qhx (J)                         | 2.451E+03 |
| Qcoils (J)                      | 2.06E+03  |
| Qsurr (J)                       | 0.00      |
| Q13 (kJ)                        | 4.51      |
| Overall Heat Trans (U)          | 1.087E-01 |
| NuT correlation                 | 1.92E+03  |
| h0_wall                         | 1.410E+05 |

Table7: Highest Impeller Speed in lab Measurements

|                      | Location | Value | Unit |  |  |  |  |
|----------------------|----------|-------|------|--|--|--|--|
| RPM of Impeller (np) | Top      | 773   | rpm  |  |  |  |  |

## Heat Transfer

University of Illinois

|                        |               |        |     |      |                  |             |              |
|------------------------|---------------|--------|-----|------|------------------|-------------|--------------|
| FlowRates              |               |        |     |      |                  |             |              |
| CW in cooling coil     | Left          | 0.3    | gpm |      | Steam Temp       | 100         | C            |
| Recirculation Water    | Right         | 0.143  | gpm |      | Steam Condensate |             |              |
| CW to HEX              | Back          | 0.7755 | gpm |      | Time (s)         | Volume (mL) | Vflow (mL/s) |
| Temperatures           |               |        |     |      | 9.81             | 22          | 2.2426       |
| WATER                  | On Vessel     | 146    | F C | 63.3 |                  |             |              |
| CW IN                  | Back left     | 38     | F C | 3.3  | Viscosity        | 0.0005      | kg/m*s       |
| CW OUT                 | Back Right    | 84     | F C | 28.9 | Cp               | 4184        | J/smK        |
| PW before HEX          | Right bott    | 149    | F C | 65.0 | Thermal K        | 0.66        | J/smK        |
| PW after HEX           | Right top     | 116.6  | F C | 47.0 |                  | (pg62)      |              |
| CW Temp IN steam trap  | Bottom        | 41     | F C | 5.0  |                  |             |              |
| CW Temp OUT steam trap | Bellow Vessel | 48     | F C | 8.9  |                  |             |              |

Table8: Highest Impeller speed Calculations

|                                 |           |
|---------------------------------|-----------|
| Nusselt Number (NuT)            | 1826.8    |
| Reynold Number (Reimp)          | 203028    |
| Prandtl Number (Pr)             | 3.17      |
|                                 |           |
| m_st (kg/s)                     | 1.35E-06  |
| Mass Rate per wet perim (Gamma) | 5.67E-07  |
| h0                              | 2.44E+05  |
| Qhx (J)                         | 3.676E+03 |
| Qcoils (J)                      | 2.02E+03  |
| Qsurr (J)                       | 0.00      |
| Q13 (kJ)                        | 5.70      |
| Overall Heat Trans (U)          | 1.288E-01 |
| NuT correlation                 | 1.90E+03  |
| h0_wall                         | 1.985E+05 |

Table9: Lowest recycle flow in lab measurements

|                      |          |        |      |  |                  |             |              |
|----------------------|----------|--------|------|--|------------------|-------------|--------------|
|                      | Location | Value  | Unit |  |                  |             |              |
| RPM of Impeller (np) | Top      | 700    | rpm  |  |                  |             |              |
| FlowRates            |          |        |      |  |                  |             |              |
| CW in cooling coil   | Left     | 0.3    | gpm  |  | Steam Temp       | 100         | C            |
| Recirculation Water  | Right    | 0.121  | gpm  |  | Steam Condensate |             |              |
| CW to HEX            | Back     | 0.7755 | gpm  |  | Time (s)         | Volume (mL) | Vflow (mL/s) |
| Temperatures         |          |        |      |  | 15.12            | 86          | 5.6878       |

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|                        |               |       |     |      |           |        |        |
|------------------------|---------------|-------|-----|------|-----------|--------|--------|
| WATER                  | On Vessel     | 153   | F C | 67.2 |           |        |        |
| CW IN                  | Back left     | 37    | F C | 2.8  | Viscosity | 0.0005 | kg/m*s |
| CW OUT                 | Back Right    | 85    | F C | 29.4 | Cp        | 4184   | J/smK  |
| PW before HEX          | Right bott    | 156.2 | F C | 69.0 | Thermal K | 0.66   | J/smK  |
| PW after HEX           | Right top     | 127.4 | F C | 53.0 |           | (pg62) |        |
| CW Temp IN steam trap  | Bottom        | 41    | F C | 5.0  |           |        |        |
| CW Temp OUT steam trap | Bellow Vessel | 47    | F C | 8.3  |           |        |        |

Table10: Lowest recycle flow calculations

|                                 |           |
|---------------------------------|-----------|
| Nusselt Number (NuT)            | 1709.9    |
| Reynold Number (Reimp)          | 183855    |
| Prandtl Number (Pr)             | 3.17      |
| m_st (kg/s)                     | 3.41E-06  |
| Mass Rate per wet perim (Gamma) | 1.44E-06  |
| h0                              | 1.79E+05  |
| Qhx (J)                         | 3.268E+03 |
| Qcoils (J)                      | 2.11E+03  |
| Qsurr (J)                       | 0.00      |
| Q13 (kJ)                        | 5.37      |
| Overall Heat Trans (U)          | 1.360E-01 |
| NuT correlation                 | 1.78E+03  |
| h0_wall                         | 1.456E+05 |

Table11: Medium Recycle flow rate in lab measurements

|                       |            |        |      |      |                  |             |              |
|-----------------------|------------|--------|------|------|------------------|-------------|--------------|
|                       | Location   | Value  | Unit |      |                  |             |              |
| RPM of Impeller       | Top        | 710    | rpm  |      |                  |             |              |
| FlowRates             |            |        |      |      |                  |             |              |
| CW in cooling coil    | Left       | 0.3    | gpm  |      | Steam Temp       | 100         | C            |
| Recirculation Water   | Right      | 0.2915 | gpm  |      | Steam Condensate |             |              |
| CW to HEX             | Back       | 0.7755 | gpm  |      | Time (s)         | Volume (mL) | Vflow (mL/s) |
| Temperatures          |            |        |      |      | 12.5             | 38          | 3.0400       |
| WATER                 | On Vessel  | 150    | F C  | 65.6 |                  |             |              |
| CW IN                 | Back left  | 38     | F C  | 3.3  | Viscosity        | 0.0005      | kg/m*s       |
| CW OUT                | Back Right | 84     | F C  | 28.9 | Cp               | 4184        | J/smK        |
| PW before HEX         | Right bott | 152.6  | F C  | 67.0 | Thermal K        | 0.66        | J/smK        |
| PW after HEX          | Right top  | 78.8   | F C  | 26.0 |                  | (pg62)      |              |
| CW Temp IN steam trap | Bottom     | 40     | F C  | 4.4  |                  |             |              |

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|------------------------|---------------|----|-----|-----|--|--|--|
| CW Temp OUT steam trap | Bellow Vessel | 48 | F C | 8.9 |  |  |  |
|------------------------|---------------|----|-----|-----|--|--|--|

Table12: medium recycle flow calculations

|                                 |             |
|---------------------------------|-------------|
| Nusselt Number (NuT)            | 1726.1      |
| Reynold Number (Reimp)          | 186482      |
| Prandtl Number (Pr)             | 3.17        |
| m_st (kg/s)                     | 1.82E-06    |
| Mass Rate per wet perim (Gamma) | 7.68E-07    |
| h0                              | 2.21E+05    |
| Qhx (J)                         | 8.373E+03   |
| Qcoils (J)                      | 2.02E+03    |
| Qsurr (J)                       | 0.00        |
| Q13 (kJ)                        | 10.39       |
| Overall Heat Trans (U)          | 0.250200322 |
| NuT correlation                 | 1797.3572   |
| h0_wall                         | 3.99E+15    |

Table13: highest recycle flow in lab measurements

|                        | Location      | Value  | Unit |      |                  |             |              |
|------------------------|---------------|--------|------|------|------------------|-------------|--------------|
| RPM of Impeller        | Top           | 710    | rpm  |      |                  |             |              |
| FlowRates              |               |        |      |      |                  |             |              |
| CW in cooling coil     | Left          | 0.3    | gpm  |      | Steam Temp       | 100         | C            |
| Recirculation Water    | Right         | 0.649  | gpm  |      | Steam Condensate |             |              |
| CW to HEX              | Back          | 0.7755 | gpm  |      | Time (s)         | Volume (mL) | Vflow (mL/s) |
| Temperatures           |               |        |      |      | 10.01            | 53          | 5.2947       |
| WATER                  | On Vessel     | 148    | F C  | 64.4 |                  |             |              |
| CW IN                  | Back left     | 37     | F C  | 2.8  | Viscosity        | 0.0005      | kg/m*s       |
| CW OUT                 | Back Right    | 82     | F C  | 27.8 | Cp               | 4184        | J/smK        |
| PW before HEX          | Right bott    | 150.8  | F C  | 66.0 | Thermal K        | 0.66        | J/smK        |
| PW after HEX           | Right top     | 140    | F C  | 60.0 |                  | (pg62)      |              |
| CW Temp IN steam trap  | Bottom        | 40     | F C  | 4.4  |                  |             |              |
| CW Temp OUT steam trap | Bellow Vessel | 46     | F C  | 7.8  |                  |             |              |

Table14: highest recycle flow calculations

|                      |        |
|----------------------|--------|
| Nusselt Number (NuT) | 1726.1 |
|----------------------|--------|

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|                                 |           |
|---------------------------------|-----------|
| Reynold Number (Reimp)          | 186482    |
| Prandtl Number (Pr)             | 3.17      |
| m_st (kg/s)                     | 3.18E-06  |
| Mass Rate per wet perim (Gamma) | 1.34E-06  |
| h0                              | 1.83E+05  |
| Qhx (J)                         | 1.225E+03 |
| Qcoils (J)                      | 1.98E+03  |
| Qsurr (J)                       | 0.00      |
| Q13 (kJ)                        | 3.20      |
| Overall Heat Trans (U)          | 7.465E-02 |
| NuT correlation                 | 1797.36   |
| h0_wall                         | 2.294E+15 |

Commented [DL36]: Unit?

Table15: Unsteady State data for low RPM

| Water<br>C<br>On<br>Vessel | Cp<br>J/s<br>mK | Visco<br>sity<br>kg/m<br>s | Ther<br>m K<br>J/sm<br>K | Temperatures                           |                                                | Total Time<br>(min) | Volume<br>(mL) | Time<br>(s) |
|----------------------------|-----------------|----------------------------|--------------------------|----------------------------------------|------------------------------------------------|---------------------|----------------|-------------|
|                            |                 |                            |                          | CW Temp IN<br>steam trap (C)<br>Bottom | CW Temp OUT<br>steam trap (C)<br>Bellow Vessel |                     |                |             |
| 25.3                       |                 |                            |                          | 4.4                                    | 4.4                                            | 0                   | 0              | 0           |
| 37.8                       | 418<br>4        | 0.000<br>7                 | 0.62<br>5                | 4.4                                    | 5.56                                           | 0.5                 | 121            | 5.07        |
| 40.6                       | 418<br>4        | 0.000<br>65                | 0.63                     | 4.4                                    | 6.67                                           | 0.87                | 109            | 5.02        |
| 43.3                       | 418<br>4        | 0.000<br>62                | 0.63<br>5                | 4.4                                    | 6.67                                           | 1.25                | 100            | 4.96        |
| 46.1                       | 418<br>4        | 0.000<br>6                 | 0.64                     | 4.4                                    | 6.67                                           | 1.55                | 103            | 5           |
| 48.9                       | 418<br>4        | 0.000<br>55                | 0.64<br>5                | 4.4                                    | 6.67                                           | 2.25                | 89             | 5.11        |
| 51.7                       | 418<br>4        | 0.000<br>5                 | 0.64<br>75               | 4.4                                    | 7.22                                           | 2.5                 | 69             | 4.94        |
| 54.4                       | 418<br>4        | 0.000<br>475               | 0.65                     | 4.4                                    | 7.22                                           | 3.17                | 72             | 5.1         |
| 57.2                       | 418<br>4        | 0.000<br>45                | 0.65<br>2                | 4.4                                    | 7.78                                           | 3.55                | 63             | 5           |
| 62.8                       | 418<br>4        | 0.000<br>425               | 0.65<br>3                | 4.4                                    | 9.44                                           | 4.49                | 49             | 4.82        |
| 65.6                       | 418<br>4        | 0.000<br>4                 | 0.66                     | 4.4                                    | 8.33                                           | 5.21                | 70             | 5.15        |
| 68.3                       | 418<br>4        | 0.000<br>375               | 0.66<br>5                | 4.4                                    | 10.00                                          | 5.48                | 77             | 4.94        |

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|      |          |              |            |     |       |      |     |      |
|------|----------|--------------|------------|-----|-------|------|-----|------|
| 71.1 | 418<br>4 | 0.000<br>35  | 0.66<br>67 | 4.4 | 11.67 | 6.18 | 86  | 4.85 |
| 73.9 | 418<br>4 | 0.000<br>325 | 0.67       | 4.4 | 13.33 | 6.45 | 102 | 4.98 |
| 76.7 | 418<br>4 | 0.000<br>31  | 0.67<br>2  | 4.4 | 14.44 | 7.2  | 107 | 4.98 |
| 79.4 | 418<br>4 | 0.000<br>3   | 0.67<br>5  | 4.4 | 11.11 | 7.54 | 20  | 4.92 |
| 82.2 | 418<br>4 | 0.000<br>3   | 0.68       | 4.4 | 10.00 | 8.3  | 32  | 9.94 |
| 85   | 418<br>4 | 0.000<br>275 | 0.68<br>5  | 4.4 | 10.00 | 9    | 32  | 9.82 |

Table16: Unsteady State data for low RPM calculations

| U        | Q13<br>kJ | Re       | Pr   | NuT      | ho     | NuT<br>correlation |
|----------|-----------|----------|------|----------|--------|--------------------|
| 10303.23 | 13.80     | 1.37E+05 | 4.69 | 1.60E+03 | 243.70 | 1.66E+03           |
| 5304.00  | 14.21     | 1.47E+05 | 4.32 | 1.64E+03 | 242.05 | 1.70E+03           |
| 3414.61  | 9.15      | 1.55E+05 | 4.09 | 1.66E+03 | 239.89 | 1.73E+03           |
| 2813.64  | 7.54      | 1.60E+05 | 3.92 | 1.67E+03 | 244.27 | 1.74E+03           |
| 1638.77  | 4.39      | 1.74E+05 | 3.57 | 1.72E+03 | 237.77 | 1.79E+03           |
| 1178.17  | 3.95      | 1.92E+05 | 3.23 | 1.77E+03 | 228.04 | 1.84E+03           |
| 939.14   | 3.15      | 2.02E+05 | 3.06 | 1.80E+03 | 232.80 | 1.87E+03           |
| 746.50   | 3.00      | 2.13E+05 | 2.89 | 1.83E+03 | 228.22 | 1.90E+03           |
| 474.11   | 2.86      | 2.26E+05 | 2.72 | 1.86E+03 | 216.55 | 1.94E+03           |
| 547.67   | 2.57      | 2.40E+05 | 2.54 | 1.89E+03 | 243.43 | 1.97E+03           |
| 595.08   | 3.99      | 2.56E+05 | 2.36 | 1.93E+03 | 260.34 | 2.01E+03           |
| 599.19   | 5.22      | 2.74E+05 | 2.20 | 1.97E+03 | 278.09 | 2.06E+03           |
| 662.38   | 7.10      | 2.95E+05 | 2.03 | 2.02E+03 | 299.08 | 2.11E+03           |
| 622.13   | 7.50      | 3.09E+05 | 1.93 | 2.05E+03 | 308.72 | 2.14E+03           |
| 112.64   | 0.91      | 3.20E+05 | 1.86 | 2.07E+03 | 179.17 | 2.16E+03           |
| 81.15    | 0.54      | 3.20E+05 | 1.85 | 2.06E+03 | 165.77 | 2.15E+03           |
| 75.75    | 0.51      | 3.49E+05 | 1.68 | 2.12E+03 | 171.34 | 2.21E+03           |

Table17: Unsteady State data for high RPM

| Water C    | Cp     | Viscosity | Therm K | CW Temp IN steam trap (C) | CW Temp OUT steam trap (C) | Total Time (min) | Volume (mL) | Time (s) | Vflow |
|------------|--------|-----------|---------|---------------------------|----------------------------|------------------|-------------|----------|-------|
| On Vesicle | J/s mK | kg/m s    | J/sm K  |                           |                            |                  |             |          |       |

|                   |          |              |            |      |       |      |     |      |                 |
|-------------------|----------|--------------|------------|------|-------|------|-----|------|-----------------|
| -<br>17.77<br>778 |          |              |            | 4.40 | 4.40  | 4.4  |     |      |                 |
| 37.77<br>7778     | 41<br>84 | 0.00<br>07   | 0.62<br>5  | 7.22 | 7.78  | 0    | 150 | 5.16 | 29.069<br>76744 |
| 40.55<br>5556     | 41<br>84 | 0.00<br>065  | 0.63       | 7.22 | 10.00 | 0.63 | 141 | 5.04 | 27.976<br>19048 |
| 43.33<br>3333     | 41<br>84 | 0.00<br>062  | 0.63<br>5  | 7.22 | 10.00 | 1.27 | 96  | 5    | 19.2            |
| 46.11<br>1111     | 41<br>84 | 0.00<br>06   | 0.64       | 7.22 | 10.00 | 2.01 | 58  | 5.07 | 11.439<br>84221 |
| 48.88<br>8889     | 41<br>84 | 0.00<br>055  | 0.64<br>5  | 7.22 | 10.56 | 2.26 | 58  | 5.28 | 10.984<br>84848 |
| 51.66<br>6667     | 41<br>84 | 0.00<br>05   | 0.64<br>75 | 7.22 | 10.56 | 3.01 | 39  | 4.95 | 7.8787<br>87879 |
| 54.44<br>4444     | 41<br>84 | 0.00<br>0475 | 0.65       | 7.22 | 10.56 | 3.33 | 40  | 5.22 | 7.6628<br>35249 |
| 57.22<br>2222     | 41<br>84 | 0.00<br>045  | 0.65<br>2  | 7.22 | 10.56 | 4.02 | 48  | 5.23 | 9.1778<br>20268 |
| 60<br>84          | 41<br>84 | 0.00<br>0438 | 0.65<br>25 | 7.22 | 10.56 | 4.22 | 39  | 4.78 | 8.1589<br>95816 |
| 60<br>84          | 41<br>84 | 0.00<br>0425 | 0.65<br>3  | 7.22 | 10.56 | 4.54 | 40  | 5.05 | 7.9207<br>92079 |
| 62.77<br>7778     | 41<br>84 | 0.00<br>04   | 0.66       | 7.22 | 10.56 | 5.23 | 48  | 5.07 | 9.4674<br>55621 |
| 65.55<br>5556     | 41<br>84 | 0.00<br>0375 | 0.66<br>5  | 7.22 | 10.56 | 5.58 | 51  | 4.98 | 10.240<br>96386 |
| 68.33<br>3333     | 41<br>84 | 0.00<br>035  | 0.66<br>67 | 7.22 | 10.56 | 6.22 | 59  | 5.01 | 11.776<br>44711 |
| 71.11<br>1111     | 41<br>84 | 0.00<br>0325 | 0.67       | 7.22 | 11.11 | 7    | 64  | 4.99 | 12.825<br>6513  |
| 73.88<br>8889     | 41<br>84 | 0.00<br>031  | 0.67<br>2  | 7.22 | 11.11 | 7.3  | 74  | 5.21 | 14.203<br>45489 |
| 76.66<br>6667     | 41<br>84 | 0.00<br>03   | 0.67<br>5  | 7.22 | 11.11 | 8.06 | 81  | 5.3  | 15.283<br>01887 |
| 79.44<br>4444     | 41<br>84 | 0.00<br>03   | 0.68       | 7.22 | 11.67 | 8.27 | 86  | 4.94 | 17.408<br>90688 |
| 82.22<br>2222     | 41<br>84 | 0.00<br>0275 | 0.68<br>5  | 5.40 | 12.22 | 9.15 | 94  | 4.89 | 19.222<br>90389 |

Table 18: Unsteady state data for high RPM calculations

| U        | Q13   | Re       | Pr   | NuT      | ho     | NuT         |
|----------|-------|----------|------|----------|--------|-------------|
|          | kJ    |          |      |          |        | correlation |
| -8337.29 | -5.59 | 1.69E+05 | 4.69 | 1.84E+03 | 260.26 | 1915.69     |

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|          |        |          |      |          |        |         |
|----------|--------|----------|------|----------|--------|---------|
| -3107.81 | -10.41 | 1.82E+05 | 4.32 | 1.88E+03 | 263.38 | 1958.88 |
| -2566.73 | -8.60  | 1.91E+05 | 4.09 | 1.91E+03 | 236.01 | 1985.05 |
| -2002.84 | -6.71  | 1.97E+05 | 3.92 | 1.92E+03 | 200.78 | 2001.85 |
| -1967.44 | -7.91  | 2.15E+05 | 3.57 | 1.97E+03 | 203.91 | 2056.02 |
| -2172.53 | -8.73  | 2.36E+05 | 3.23 | 2.03E+03 | 188.42 | 2120.32 |
| -2744.90 | -11.03 | 2.49E+05 | 3.06 | 2.07E+03 | 189.90 | 2154.49 |
| -9257.14 | -37.21 | 2.63E+05 | 2.89 | 2.10E+03 | 205.34 | 2191.82 |
| #####    | -69.83 | 2.70E+05 | 2.81 | 2.12E+03 | 199.30 | 2212.14 |
| 21685.11 | 87.17  | 2.78E+05 | 2.72 | 2.14E+03 | 199.26 | 2233.26 |
| 4371.96  | 17.57  | 2.95E+05 | 2.54 | 2.18E+03 | 215.78 | 2271.22 |
| 3326.44  | 13.37  | 3.15E+05 | 2.36 | 2.22E+03 | 226.32 | 2315.28 |
| 2480.07  | 9.97   | 3.38E+05 | 2.20 | 2.27E+03 | 242.63 | 2367.67 |
| 1766.88  | 8.29   | 3.64E+05 | 2.03 | 2.32E+03 | 255.87 | 2423.50 |
| 1754.27  | 8.23   | 3.81E+05 | 1.93 | 2.36E+03 | 268.93 | 2459.92 |
| 1495.65  | 7.01   | 3.94E+05 | 1.86 | 2.38E+03 | 278.61 | 2483.53 |
| 1526.55  | 8.18   | 3.94E+05 | 1.85 | 2.37E+03 | 290.97 | 2477.43 |
| 963.08   | 7.92   | 4.30E+05 | 1.68 | 2.44E+03 | 309.59 | 2544.85 |

**8. Appendix II: Error Analysis**

There happened to be some errors in this lab where error could have taken place. The impeller speed fluctuating quite drastically during the trials that were performed. This fluctuation could have altered the data and analysis slightly, since it wasn't completely steady during the experiments. Also, to measure the volume flow rate of the steam trap, a stopwatch and a graduated cylinder was used. It wasn't completely accurate since it was hard to stop collecting the fluid and stop the timer at the exact same time. This could have caused some error in the data that was collected.

**Commented [DL37]:** 8/10

Quantitative analysis is needed. Like the comparison of the values of the dimensionless groups calculated from the empirical equation and obtained from the experiment, or the uncertainty of the readout.

## 9. Appendix III: Sample Calculations

Commented [DL38]: 10/10  
Which sample?

- 1.
- Impeller Reynolds number:

$$Re_{imp} = \frac{D_I^2 n \rho}{\mu}$$

When impeller speed is 610 rpm and temperature of water in tank is 67.8°C:

$$Re_{imp} = \frac{(0.0889 \text{ m})^2 \left(\frac{610 \text{ rpm}}{60}\right) \left(997 \frac{\text{kg}}{\text{m}^3}\right)}{0.0005 \frac{\text{kg}}{\text{m} \cdot \text{s}}} = \mathbf{160,217}$$

- 2.
- Fluid Prandtl number:

$$Pr = \frac{C_p \mu}{k}$$

When impeller speed is 610 rpm and temperature of water in tank is 67.8°C:

$$Pr = \frac{\left(4184 \frac{\text{J}}{\text{smK}}\right) \left(0.0005 \frac{\text{kg}}{\text{m} \cdot \text{s}}\right)}{0.66 \text{ J/smK}} = \mathbf{3.17}$$

- 3.
- Heat transfer coefficient correlation to tank Nusselt number
- for an unbaffled vessel containing Newtonian fluid:

$$Nu_T = \frac{h_i D_T}{k} = 0.36 \left(\frac{D_I^2 n \rho}{\mu}\right)^{\frac{2}{3}} \left(\frac{C_p \mu}{k}\right)^{\frac{1}{3}}$$

$h_i$  = convective heat transfer coefficient at the inner surface of the tank

$D_T$  = tank diameter

$\mu$  = viscosity of the fluid

$C_p$  = specific heat of the fluid

$k$  = thermal conductivity of the fluid

$\rho$  = density of the fluid

$D_I$  = impeller diameter

$n$  = speed of rotation of the impeller, (revolutions per unit time)

Dimensionless Nusselt number when impeller speeds is 610 rpm and temperature of water in tank is 67.8°C:

$$Nu_T = 0.36 \left(\frac{D_I^2 n \rho}{\mu}\right)^{\frac{2}{3}} \left(\frac{C_p \mu}{k}\right)^{\frac{1}{3}} = 0.36 (Re)^{\frac{2}{3}} (Pr)^{\frac{1}{3}}$$

$$Nu_T = 0.36 (160,217)^{\frac{2}{3}} (3.17)^{\frac{1}{3}} = \mathbf{1560}$$

4. Mass rate of steam condensation:

$$\Gamma = \frac{\dot{m}_{st}}{\pi D_{T,o}}$$

When impeller speed is 610 rpm and temperature of water in tank is 67.8°C:

$$\Gamma = \frac{2.5 * 10^{-6} \text{ kg/s}}{\pi(0.7556 \text{ m})} = 1.06 * 10^{-6} \frac{\text{kg}}{\text{m} * \text{s}}$$

5. Heat transfer coefficient for saturated steam at 1 atm pressure:

$$h_o = 2960 \left( \frac{D_{T,o}}{\dot{m}_{st}} \right)^{\frac{1}{3}}$$

When impeller speed is 610 rpm and temperature of water in tank is 67.8°C:

$$h_o = 2960 \left( \frac{0.7556 \text{ m}}{2.54 * 10^{-6} \text{ kg/s}} \right)^{\frac{1}{3}} = 1.98 * 10^5 \frac{\text{m} * \text{s}}{\text{kg}}$$

6. Example of heat flux through heat exchanger:

$$Q_{HX} = \left[ \left( \frac{n}{60} \right) \right] \rho C_p (t_{HX \text{ inlet water}} - t_{HX \text{ outlet water}})$$

When impeller speed is 610 rpm and temperature of water in tank is 67.8°C:

$$Q_{HX} = \left[ \left( \frac{610 \text{ rpm}}{60} \right) \right] \left( 997 \frac{\text{kg}}{\text{m}^3} \right) * (69.5^\circ\text{C} - 49.5^\circ\text{C}) = 4.085 * 10^3 \text{ J}$$

7. Overall heat transfer coefficient:

$$U_{i,\text{wall}} = \frac{Q_{13}}{(t_1 - t_3)A} = \frac{(Q_{HX} + Q_{\text{coils}} + Q_{\text{surr}})/1000}{(t_1 - t_3)A}$$

When impeller speed is 610 rpm and temperature of water in tank is 67.8°C:

$$U_{i,\text{wall}} = \frac{(4.085 * 10^3 \text{ J} + 2.02 * 10^3 \text{ J} + 0 \text{ J})/1000}{(100^\circ\text{C} - 67.8^\circ\text{C})(1.21 \text{ m}^2)} = 0.1571 \frac{\text{kJ}}{^\circ\text{C} * \text{m}^2}$$

8. Nu<sub>T</sub> correlation for h<sub>i,wall</sub>:

$$\frac{\left(\frac{hD_T}{k}\right)}{\left(\frac{C_p\mu}{k}\right)^{0.33} \left(\frac{\mu}{\mu_w}\right)^{0.14}} = 0.36 \left(\frac{D_T^2 n \rho}{\mu}\right)^{0.67} = \log \left[ \frac{Nu_T}{Pr^{\frac{1}{3}}} \right] = \log(0.36) + 0.67 \log [Re_I]$$

When impeller speed is 610 rpm and temperature of water in tank is 67.8°C:

$$Nu_T = 10^{\log(0.36) + 0.67 \log(160,217)} * (3.17)^{\frac{1}{3}} = \mathbf{1.62 * 10^3}$$

9. Heat transfer coefficient on the outside of the vessel wall correlation:

$$\frac{h_o L_T}{k} = 0.925 \left( \frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{\frac{1}{3}}$$

$$\therefore h_{o,wall} = 0.925 \left( \frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{\frac{1}{3}} \left( \frac{k}{L_T} \right)$$

When impeller speed is 610 rpm and temperature of water in tank is 67.8°C:

$$h_{o,wall} = 0.925 \left( \frac{(0.508 \text{ m})^3 \left(997 \frac{\text{kg}}{\text{m}^3}\right)^2 \left(9.8 \frac{\text{m}}{\text{s}^2}\right)}{\left(0.0005 \frac{\text{kg}}{\text{m} \cdot \text{s}}\right) (1.07 * 10^6)} \right)^{\frac{1}{3}} \left( \frac{0.66 \frac{\text{J}}{\text{smK}}}{0.508 \text{ m}} \right) = \mathbf{1.61 * 10^5 \frac{W}{m^2 K}}$$



**10. Appendix IV: Individual Team Contributions**

NAME: Jake Proctor

| Jake Proctor                              | <b>TIME<br/>(HOURS)</b> | <b>DESCRIPTION</b>               |
|-------------------------------------------|-------------------------|----------------------------------|
| <b>Operator (Both Lab Days)</b>           | 5                       | Prelab and in lab experiment     |
| <b>Pre-Lab Editing</b>                    | 0.25                    | Spelling checks                  |
| <b>Experimental Objective (Prelab)</b>    | 0                       |                                  |
| <b>Apparatus (prelab)</b>                 | 0                       |                                  |
| <b>Materials &amp; Supplies (prelab)</b>  | 0.25                    | Wrote and filled out section     |
| <b>Procedure (prelab)</b>                 | 0.5                     | Wrote procedures                 |
| <b>Project Safety Evaluation (prelab)</b> | 0.25                    | Filled out sheet                 |
| <b>Final Lab Editing</b>                  | 0.25                    | Spelling checks and organization |
| <b>Executive Summary (final lab)</b>      | 0                       |                                  |
| <b>Introduction (Final lab)</b>           | 0                       |                                  |
| <b>Theory (Final Lab)</b>                 | 5                       | Researched and wrote theory      |
| <b>Results (final lab)</b>                | 0                       |                                  |
| <b>Discussion (final lab)</b>             | 0                       |                                  |
| <b>References (final and/or prelab)</b>   | 0.5                     | Added 4 references               |
| <b>Data Tabulation/Graphs (final)</b>     | 0                       |                                  |
| <b>Error Analysis (final lab)</b>         | 0                       |                                  |
| <b>Sample Calculations (final Lab)</b>    | 0                       |                                  |
| <b>PowerPoint Presentation</b>            | 1                       | Created PowerPoint               |
| <b>Total</b>                              | 13.25                   |                                  |

NAME: Courteney Fabbri

|                                    | TIME (HOURS) | DESCRIPTION                              |
|------------------------------------|--------------|------------------------------------------|
| Operator (Both Lab Days)           | 1            | Timed condensate collection              |
| Pre-Lab Editing                    | 0            |                                          |
| Experimental Objective (Prelab)    | 0            |                                          |
| Apparatus (prelab)                 | 1            | Labeled and wrote apparatus descriptions |
| Materials & Supplies (prelab)      | 0            |                                          |
| Procedure (prelab)                 | 0            |                                          |
| Project Safety Evaluation (prelab) | 0            |                                          |
| Final Lab Editing                  | 0            |                                          |
| Executive Summary (final lab)      | 2            |                                          |
| Introduction (Final lab)           | 0            |                                          |
| Theory (Final Lab)                 | 0            |                                          |
| Results (final lab)                | 0            |                                          |
| Discussion (final lab)             | 0            |                                          |
| References (final and/or prelab)   | 0            |                                          |
| Data Tabulation/Graphs (final)     | 0            |                                          |
| Error Analysis (final lab)         | 0            |                                          |
| Sample Calculations (final Lab)    | 3            |                                          |
| Power Point Presentation           | 0            |                                          |
| Total                              | 7            |                                          |

NAME: Erica Caruso

|                                    | TIME (HOURS) | DESCRIPTION |
|------------------------------------|--------------|-------------|
| Operator (Both Lab Days)           | 5            |             |
| Pre-Lab Editing                    | 0.25         |             |
| Experimental Objective (Prelab)    | 0.25         |             |
| Apparatus (prelab)                 | 0            |             |
| Materials & Supplies (prelab)      | 0            |             |
| Procedure (prelab)                 | 0            |             |
| Project Safety Evaluation (prelab) | 0            |             |
| Final Lab Editing                  | 0.25         |             |
| Executive Summary (final lab)      | 0            |             |
| Introduction (Final lab)           | 1            |             |
| Theory (Final Lab)                 | 0            |             |
| Results (final lab)                | 0            |             |
| Discussion (final lab)             | 0            |             |
| References (final and/or prelab)   | 0.25         |             |
| Data Tabulation/Graphs (final)     | 0            |             |
| Error Analysis (final lab)         | 0.5          |             |
| Sample Calculations (final Lab)    | 0            |             |
| Power Point Presentation           | 0            |             |
| Total                              | 7.5          |             |

NAME: Milos Markovic

|                                    | TIME (HOURS) | DESCRIPTION |
|------------------------------------|--------------|-------------|
| Operator (Both Lab Days)           | 4            |             |
| Pre-Lab Editing                    | 0            |             |
| Experimental Objective (Prelab)    | 0            |             |
| Apparatus (prelab)                 | 0            |             |
| Materials & Supplies (prelab)      | 0            |             |
| Procedure (prelab)                 | 0            |             |
| Project Safety Evaluation (prelab) | 0            |             |
| Final Lab Editing                  | 1            |             |
| Executive Summary (final lab)      | 0            |             |
| Introduction (Final lab)           | 0            |             |
| Theory (Final Lab)                 | 0            |             |
| Results (final lab)                | 6            |             |
| Discussion (final lab)             | 0            |             |
| References (final and/or prelab)   | 0            |             |
| Data Tabulation/Graphs (final)     | 2            |             |
| Error Analysis (final lab)         | 0            |             |
| Sample Calculations (final Lab)    | 0            |             |
| Power Point Presentation           | 0            |             |
| Total                              | 13           |             |

NAME: Mike Simon

|                                    | TIME (HOURS) | DESCRIPTION     |
|------------------------------------|--------------|-----------------|
| Operator (Both Lab Days)           | 5            | Data collection |
| Pre-Lab Editing                    |              |                 |
| Experimental Objective (Prelab)    |              |                 |
| Apparatus (prelab)                 | 0.5          |                 |
| Materials & Supplies (prelab)      |              |                 |
| Procedure (prelab)                 |              |                 |
| Project Safety Evaluation (prelab) |              |                 |
| Final Lab Editing                  | 0.5          |                 |
| Executive Summary (final lab)      |              |                 |
| Introduction (Final lab)           |              |                 |
| Theory (Final Lab)                 |              |                 |
| Results (final lab)                |              |                 |
| Discussion (final lab)             | 1            |                 |
| References (final and/or prelab)   |              |                 |
| Data Tabulation/Graphs (final)     |              |                 |
| Error Analysis (final lab)         |              |                 |
| Sample Calculations (final Lab)    |              |                 |
| Power Point Presentation           |              |                 |
| Total                              | 7            |                 |